

Termo IB - Resumen de fórmulas

1. Gases ideales

$$PV = nRT \quad PM = \rho RT, \quad \rho \equiv \text{densidad}, \mathcal{M} \equiv \text{Masa molar} \quad (1)$$

2. Conservación de la energía

$$dE = dU + \sum dE_i = \delta Q - \delta W \quad (2)$$

$$dU = nc_v dT \quad dH = dU + d(PV) \quad (3)$$

$$\text{Si } P = \text{cte}, \quad dH = dU + PdV = nc_p dT \quad (4)$$

3. Transformaciones adiabáticas cuasiestáticas en gases

$$c_p = \{3/2R, 5/2R, 3R\} \quad c_v = c_p + R \quad (5)$$

$$\gamma = \frac{c_p}{c_v} \quad TV^{\gamma-1} = c_1; TP^{(1-\gamma)/\gamma} = c_2; PV^\gamma = c_3; \quad (6)$$

4. Sistemas abiertos

$$h^0 = h + gz + \frac{1}{2}c^2 \quad (7)$$

$$dE_{VC} = \delta Q + \sum_k h_{in}^0 |\delta m| - \sum_k h_{out}^0 |\delta m| - \sum_k \delta W_k - \delta W_{ExpVC} \quad (8)$$

$$\delta m_{VC} = \sum_k |\delta m_{in}| - \sum_k |\delta m_{out}| \quad (9)$$

$$\text{Bernoulli: } P_2 + \rho g z_2 + \frac{1}{2} \rho c_2^2 = P_1 + \rho g z_1 + \frac{1}{2} \rho c_1^2 = \text{cte} \quad (10)$$

5. Dispositivos en régimen estacionario

$$\text{Bomba (líquido)} - w_{eje} = h_2 - h_1 = \Delta u + v \Delta P \quad (11)$$

$$\text{Turbina: } -w_{eje} = \Delta H = c_p \Delta T \quad (12)$$

$$\text{Intercambiador de calor: } \sum_k h_{out}^0 \dot{m}_{out} = \sum_k h_{in}^0 \dot{m}_{in} \quad (13)$$

$$(14)$$

Tobera-Difusor:

$$h_1^+ = h_1 + \frac{1}{2} c_1^2 = h_2^+ \quad \dot{m} = c_1 A_1 \rho_1 = c_2 A_2 \rho_2 = \text{cte} \quad (15)$$

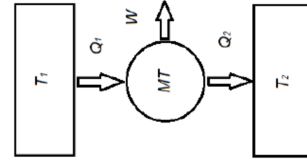
6. Segundo principio

$$\text{Máquina Térmica: } W = Q_1 - Q_2 \quad (16)$$

$$\eta_{MT} = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1} \Rightarrow 1 - \frac{T_2}{T_1} \quad (17)$$

$$\text{Máquina Frigorífica: } Q_1 = Q_2 + W \quad \beta_{MF} = \frac{Q_2}{W} \quad (18)$$

$$\beta_{BC} = \frac{Q_1}{W} = \frac{Q_2 + W}{W} \quad (19)$$



$$\text{Entropía: } dS = \frac{\delta Q_{rev}}{T} \quad dS_{Real} \geq 0 \quad dS = \frac{\delta Q}{T} + dS_{gen} \quad (20)$$

$$ds = \frac{du}{T} + \frac{Pdv}{T} = \frac{dh}{T} - \frac{vdP}{T} \quad (21)$$

$$\text{Líquidos/Sólidos incompresibles: } c_v = c_p = C, dv = 0 \quad (22)$$

$$ds = \frac{CdT}{T} \quad (23)$$

$$\text{Gases ideales: } ds = \frac{c_v dT}{T} + \frac{Rdv}{v} \quad (24)$$

6.1. Sistemas abiertos con j entradas y salidas y k fuentes de temperatura constante:

$$dS_{VC} \geq \sum_j \delta m_j + \sum_k \frac{\delta Q_k}{T_k} \quad (25)$$

$$\text{Escrito por unidad de tiempo en régimen estacionario:} \quad (26)$$

$$-\sum_j s_j \dot{m}_j = \sum_k \frac{\dot{Q}_k}{T_k} + \dot{S}_{gen} \quad (27)$$

$$\text{Trabajo para una máquina en flujo estable:} \quad (28)$$

$$w_{rev} = - \int_1^2 v dP - \Delta k_E - \Delta p_E \quad (29)$$

6.2. Eficiencias isoentrópicas

$$\eta_{Bomba}^S = \frac{W_{Rev}}{W_{Real}} = \frac{v \Delta P}{c \Delta T + v \Delta P} \quad (30)$$

$$\eta_{Compresor}^S = \frac{W_{Rev}}{W_{Real}} = \frac{h_{out}^s - h_{in}}{h_{out} - h_{in}} \quad (31)$$

$$\text{Gases Ideales Perf.: } \eta_{Compresor}^S = \frac{T_{out}^s - T_{in}}{T_{out} - T_{in}} \quad (32)$$

Donde T_{out}^s se obtiene mediante las relaciones de Poisson.

$$\eta_{Tobera}^S = \left(\frac{c_{out}}{c_{out}^s} \right)^2 = \frac{h_{out}^+ - h_{out}}{h_{out}^+ - h_{out}^s} \quad (33)$$

7. Lagrangiano del modelo estándar

$$\begin{aligned}
\mathcal{L} = & -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{4}\text{tr}(\mathbf{W}_{\mu\nu}\mathbf{W}^{\mu\nu}) - \frac{1}{4}\text{tr}(\mathbf{G}_{\mu\nu}\mathbf{G}^{\mu\nu}) && \text{(gauge terms: U(1), SU(2), SU(3))} \\
& + (\bar{\nu}_L, \bar{e}_L) \tilde{\sigma}^\mu i D_\mu \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} + \bar{e}_R \sigma^\mu i D_\mu e_R + \bar{\nu}_R \sigma^\mu i D_\mu \nu_R + (\text{h.c.}) && \text{(lepton kinetic terms)} \\
& - \frac{\sqrt{2}}{\nu} \left[(\bar{\nu}_L, \bar{e}_L) \phi M^e e_R + \bar{e}_R M^{e\dagger} \phi^\dagger \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} \right] && \text{(charged lepton mass term)} \\
& - \frac{\sqrt{2}}{\nu} \left[(-\bar{e}_L, \bar{\nu}_L) \phi^* M^\nu \nu_R + \bar{\nu}_R M^{\nu\dagger} \phi^T \begin{pmatrix} -e_L \\ \nu_L \end{pmatrix} \right] && \text{(neutrino mass term)} \\
& + (\bar{u}_L, \bar{d}_L) \tilde{\sigma}^\mu i D_\mu \begin{pmatrix} u_L \\ d_L \end{pmatrix} + \bar{u}_R \sigma^\mu i D_\mu u_R + \bar{d}_R \sigma^\mu i D_\mu d_R + (\text{h.c.}) && \text{(quark kinetic terms)} \\
& - \frac{\sqrt{2}}{\nu} \left[(\bar{u}_L, \bar{d}_L) \phi M^d d_R + \bar{d}_R M^{d\dagger} \phi^\dagger \begin{pmatrix} u_L \\ d_L \end{pmatrix} \right] && \text{(down-type quark mass term)} \\
& - \frac{\sqrt{2}}{\nu} \left[(-\bar{d}_L, \bar{u}_L) \phi^* M^u u_R + \bar{u}_R M^{u\dagger} \phi^T \begin{pmatrix} -d_L \\ u_L \end{pmatrix} \right] && \text{(up-type quark mass term)} \\
& + (D_\mu \phi)^\dagger D^\mu \phi - \frac{m_h^2}{2\nu^2} \left(\phi^\dagger \phi - \frac{\nu^2}{2} \right)^2 && \text{(Higgs kinetic + potential)}
\end{aligned}$$